

From: Saravanan KRISHNAN <krishnan.saravanan@se.com>
Sent: 24 July 2026 11:30
To: Amit Jadhav; Ruchy Bhutani
Cc: Vinoth C; HEMABHUSHANAM KUPPUSWAMY; Sareesh Thiagarajan; Ravindra N KULKARNI; Srinivasan M K; Suresh G; K Santhosh REDDY; kannanthiru0712@gmail.com; Ashwini Pandey; Dinesh Kumar; ev@electrovast.com; csq@electrovast.com; Mayur Kanchan; R Saravanan; Sathishkumar Manigandan; Danish Asrar; Arun Raj; T S Narasiman; Anbukkaniraja K
Subject: RE: Reg:- CC202526C13E - ACB Breaker IG Fault Trip 3.1

Hi Amit,

Thank you for your patience, and apologies for the delay in responding to your email.

We were conducting a detailed analysis of the returned breaker and reviewing the Power Quality Analyzer (PQA) records to thoroughly investigate the trip event and address your concerns. The analysis has now been completed, and our findings are summarized below.

1. Breaker and Micrologic Analysis

The returned breaker and Micrologic trip unit were subjected to a comprehensive inspection and testing. All measured parameters were found to be within specification.

No Problem Found (NFP) was identified with either the breaker or the Micrologic trip unit.

The following checks were carried out successfully:

1. Current sensor resistance was verified and found to be healthy.
2. Current sensor wiring and connections between the current sensors and the Micrologic trip unit were checked and found to be healthy.
3. Micrologic primary current injection tests were completed successfully.
4. Micrologic secondary current injection tests were completed successfully.

Based on the above tests, there is no evidence of any malfunction in the breaker, current sensors, Current sensors wiring, or Micrologic trip unit.

2. PQA Data Analysis

The PQA records corresponding to the trip event that occurred on 23-May-2025 at 13:34:02.838 were analysed in detail.

Normal Operating Condition (Green Boxes)

The records at 13:34:02.088 and 13:34:02.338 show stable system operation:

- Voltage:
 - R Phase: 658 V
 - Y Phase: 651 V
 - B Phase: 662 V
- Current:
 - Approximately 2.42 kA on all three phases
- Total Active Power:
 - Approximately 2.758 MW
- Voltage and current phase relationships were stable in both samples recorded 250 ms apart.

These two consecutive samples confirm that the system was operating normally prior to the disturbance.

3. System Disturbance (Red Box)

At 13:34:02.588, a significant disturbance was recorded.

Voltage

The phase-to-phase voltages dropped to:

- R Phase: 573.3 V
- Y Phase: 595.5 V
- B Phase: 588.7 V

Current

Current reduced from approximately 2.42 kA to:

- R Phase: 1.717 kA
- Y Phase: 1.709 kA
- B Phase: 1.709 kA

Active Power

Total active power reduced from:

- 2.758 MW to 1.379 MW

Phase Angle Changes

A substantial change in phase angles was observed.

Voltage phase angles changed to:

- R Phase: 0°
- Y Phase: -117.7°
- B Phase: -232.0°

Current phase angles changed to:

- R Phase: -160.1°
- Y Phase: -118.5°
- B Phase: -228.3°

The most significant change occurred on the R-phase current, which shifted from approximately -4.7° to -160°.

This represents a major disturbance in the system and indicates a significant change in the electrical conditions immediately before the breaker trip.

4. Breaker Trip Confirmation (Yellow Box)

At **13:34:02.838**, the PQA records indicate:

Voltage

- R Phase: 21.15 V
- Y Phase: 20.8 V
- B Phase: 15.65 V

Current

- Near zero on all phases

Active Power

- 0 MW

These values clearly indicate that the breaker had opened and isolated the circuit.

Since the PQA meter is connected on the load side of the breaker, the presence of voltage and current during the disturbance confirms that the breaker was still in the closed position at the time of the phase-angle shift. The subsequent collapse of voltage and current confirms operation of the breaker.

5. Correlation with Ig Protection Operation

The PQA sampling interval is 250 ms.

The event timeline is as follows:

Event	Time
Normal condition	13:34:02.338

Disturbance detected	13:34:02.588
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Breaker open	13:34:02.838
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The observed sequence is consistent with the configured Ig protection delay and breaker operating time.

For a 3-pole breaker, Ig (Ground Fault) protection is based on the vector sum of the three phase currents. Under normal conditions, the vector sum should be approximately zero. When a significant phase-angle displacement occurs, the vector sum is no longer zero and a residual current is produced. This residual current is interpreted by the Micrologic as a ground fault condition.

The replacement Micrologic unit has also recorded ground current values, which further supports that the breaker trip was initiated by the Ig protection function.

Based on:

- Breaker test results,
- Micrologic event records,
- Residual current records, and
- PQA data analysis,

the evidence indicates that the breaker operated on a genuine Ig protection event and not due to nuisance tripping or equipment malfunction.

6. Conclusion

Based on the testing and analysis performed, we conclude that:

- The breaker and Micrologic trip unit are healthy.
- No equipment defect has been identified.
- The PQA data captured a system disturbance involving voltage depression and significant phase-angle shift.
- The disturbance resulted in residual current detection by the Ig protection function.
- The breaker operated within its expected protection characteristics and opening time.
- The trip is consistent with a valid Ig protection operation and is not indicative of breaker malfunction.

Subject to consultant approval, one possible solution would be to disable/bypass the Ig protection function, which would prevent similar ground-fault-related trips caused by these system disturbances.

We trust the above analysis addresses your concerns. Please let us know if you require any further clarification

Date	Function	Duration	AN(V) / A(A) Avg	BN(V) / B(A) Avg	CN(V) / C(A) Avg	NG(V) / N(A) Avg
23-05-2026 13:34:02 88msec	Vrms ph-ph		658.25 V (f)	651.2 V (f)	662.05 V (f)	
23-05-2026 13:34:02 88msec	Arms		2.44 kA (f)	2.421 kA (f)	2.422 kA (f)	
23-05-2026 13:34:02 88msec	Active Power		0.933 MW (f)	910.3 kW (f)	915 kW (f)	
23-05-2026 13:34:02 88msec	Voltage Phi		0 °	-120.16 °	-239.26 °	
23-05-2026 13:34:02 88msec	Current Phi		-4.7 °	-124.52 °	-244.64 °	
23-05-2026 13:34:02 338msec	Vrms ph-ph		658.35 V (f)	651.3 V (f)	662.1 V (f)	
23-05-2026 13:34:02 338msec	Arms		2.439 kA (f)	2.419 kA (f)	2.422 kA (f)	
23-05-2026 13:34:02 338msec	Active Power		0.933 MW (f)	910 kW (f)	915.2 kW (f)	
23-05-2026 13:34:02 338msec	Voltage Phi		0 °	-120.16 °	-239.26 °	
23-05-2026 13:34:02 338msec	Current Phi		-4.7 °	-124.54 °	-244.66 °	
23-05-2026 13:34:02 588msec	Vrms ph-ph		573.3 V (U,f)	595.5 V (U,f)	588.65 V (U,f)	
23-05-2026 13:34:02 588msec	Arms		1.717 kA (U,f)	1.709 kA (U,f)	1.709 kA (U,f)	
23-05-2026 13:34:02 588msec	Active Power		0.467 MW (U,f)	456 kW (U,f)	456.5 kW (U,f)	
23-05-2026 13:34:02 588msec	Voltage Phi		0 °	-117.66 °	-232.04 °	
23-05-2026 13:34:02 588msec	Current Phi		-160.06 °	-118.5 °	-228.3 °	
23-05-2026 13:34:02 838msec	Vrms ph-ph		21.15 V (U,f)	20.8 V (U,f)	15.65 V (U,f)	
23-05-2026 13:34:02 838msec	Arms		0.003 kA (U,f)	0.005 kA (U,f)	0.004 kA (U,f)	
23-05-2026 13:34:02 838msec	Active Power		0 MW (U,f)	0 kW (U,f)	0 kW (U,f)	
23-05-2026 13:34:02 838msec	Voltage Phi		0 °	-177.74 °	-352.02 °	
23-05-2026 13:34:02 838msec	Current Phi		-3.42 °	-0.04 °	-179.94 °	
23-05-2026 13:34:03 88msec	Vrms ph-ph		4.05 V (U,f)	4 V (U,f)	0.6 V (U,f)	
23-05-2026 13:34:03 88msec	Arms		0.000 kA (U,f)	0.002 kA (U,f)	0.000 kA (U,f)	
23-05-2026 13:34:03 88msec	Active Power		0 MW (U,f)	0 kW (U,f)	0 kW (U,f)	
23-05-2026 13:34:03 88msec	Voltage Phi		0 °	-179.34 °	-358.72 °	
23-05-2026 13:34:03 88msec	Current Phi		-199.1 °	-185.68 °	-230.08 °	

With Warm Regards,

Saravanan Krishnan

Technical Support and Enablement Leader–Services

Schneider Electric IT Business India Private Ltd

M +91 7795813931

E krishnan.saravanan@se.com

Argon North Tower,
Avinya Campus Bagmane Solarium City,
Kundalahalli Road, Doddanekundi Village,
K R Puram Hobli , Bengaluru – 560037,
Karnataka, India..



General

From: Amit Jadhav <amit.jadhav@essensrenewable.com>

Sent: 20 July 2026 16:08

To: Saravanan KRISHNAN <krishnan.saravanan@se.com>; Ruchy Bhutani <ruchy.bhutani@se.com>

Cc: Vinoth C <Vinoth.C@se.com>; HEMABHUSHANAM KUPPUSWAMY <Hemabhushanam.Kuppuswamy@se.com>;

Sareesh Thiagarajan <Sareesh.t@se.com>; Ravindra N KULKARNI <ravindra.kulkarni@se.com>; Srinivasan M K

<Srinivasan.MK@se.com>; Suresh G <suresh.g@se.com>; K Santhosh REDDY <Santhosh.Reddy@se.com>;

kannanthiru0712@gmail.com; Ashwini Pandey <ashwini.pandey@essensrenewable.com>; Dinesh Kumar

<dinesh.kumar5@se.com>; ev@electrovast.com; csq@electrovast.com; Mayur Kanchan

<mayur.kanchan@essensrenewable.com>; R Saravanan <saravanan@electrovast.com>; Sathishkumar Manigandan

<sathishkumar.mani@essensrenewable.com>; Danish Asrar <danish.asrar@essensrenewable.com>; Arun Raj

<Arun.Raj@se.com>; T S Narasiman <narasiman.sundaramoorthy@se.com>; Anbukkaniraja K

<anbukkaniraja.k@essensrenewable.com>

Subject: RE: Reg:- CC202526C13E - ACB Breaker IG Fault Trip 3.1

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